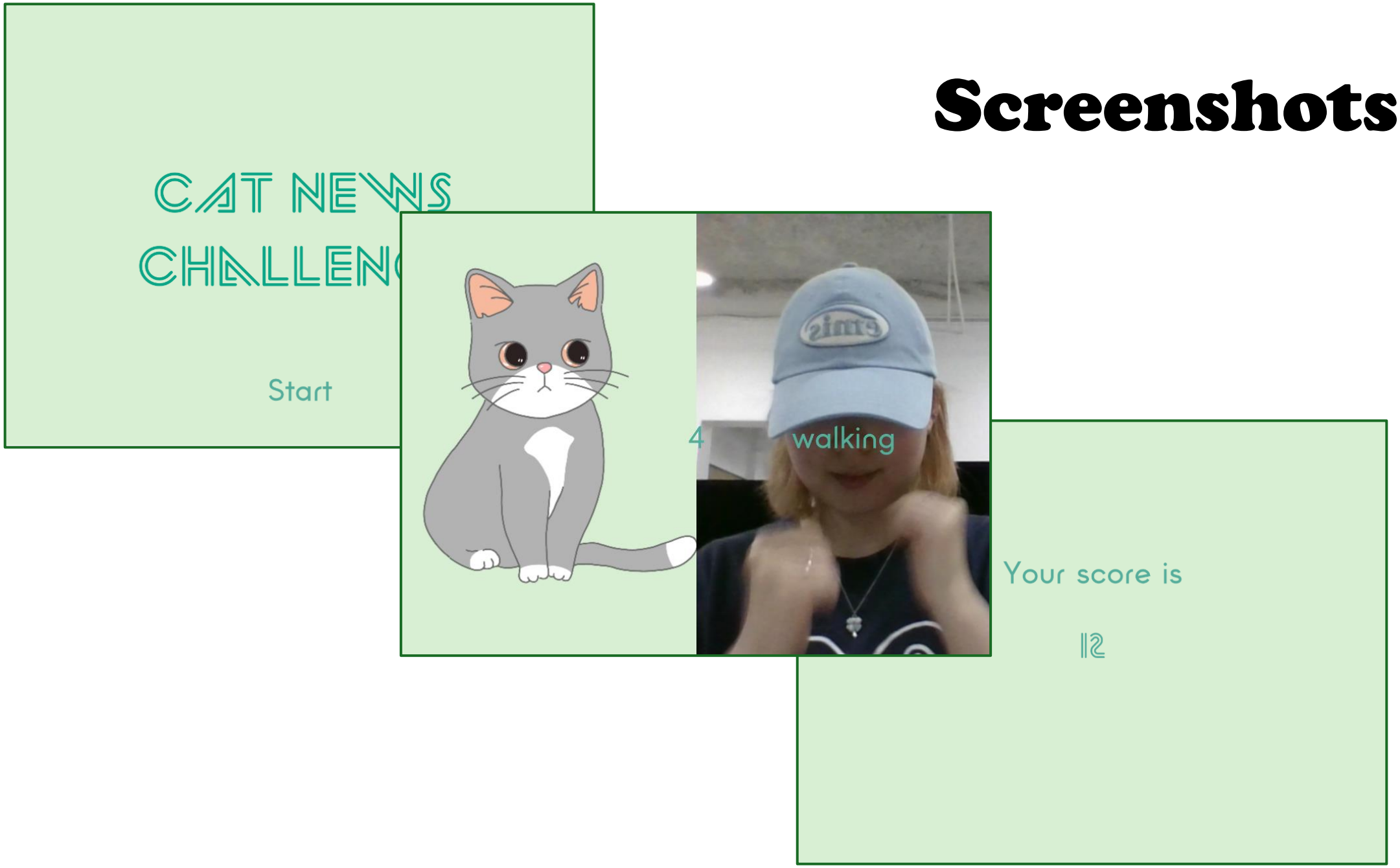
CAT NEWS Challenge

2024 Spring [AAT2004]

Intro to Creative Computing

20231005 Siyun Kim

Screenshots



Links

- Program Links
  - Fullscreen : https://editor.p5js.org/tldbs37/full/tPsvXjNDi
  - Edit : https://editor.p5js.org/tldbs37/sketches/tPsvXjNDi

- Video Link
  - https://drive.google.com/file/d/1ZmyeconM9QnqX3MiYQHp7Kpi3MIExR8k/view?usp=sharing

Motivation & How to play

- MOTIVATION
  - There are a lot of ppl who takes a short form of themselves.
  - I think short form is good medium to show themselves.
  - Especially as CAT NEWS was a big issue these days, I made a challenge game which enable ppl to make challenge short form easily.

- WHAT & HOW
  - When user clicks start button, a cat doing CAT NEWS challenge and a video via camera are shown.
  - Doing a challenge with a cute cat, user can get score.
  - The more accurate the motion is, the higher the score gets.

References

- Pose detection with PoseNet
  - Making a ring above the user's head

  https://jeeyoonhyun.netlify.app/posts/p5js-posenet-tutorial/
  - ID number of each body part

  https://www.tensorflow.org/lite/examples/pose_estimation/overview?hl=ko

- Flip camera horizontally
  - ChatGPT

```
push();
// to flip camera horizontally
translate(width, 0);
scale(-1, 1);
image(video, -350, 0); // draw video of camera
pop();
```

ID	파트
0	코
1	왼쪽 눈
2	오른쪽 눈
3	왼쪽 귀
4	오른쪽 귀
5	왼쪽 어깨
6	오른쪽 어깨
7	왼쪽 팔꿈치
8	오른쪽 팔꿈치
9	왼쪽 손목
10	오른쪽 손목
11	왼쪽 골반 부위
12	오른쪽 골반 부위
13	왼쪽 무릎
14	오른쪽 무릎
15	왼쪽 발목
16	오른쪽 발목

Code – setup()

Video settings

Load PoseNet

If video ends

If video ends, change scene1 to scene2

```
41 function setup() {
42   createCanvas(800, 600);
43   background('#D8EFD3');
44
45   catnews_video = createVideo('cat news.mp4');
46   catnews_video.size(600, 600);
47   catnews_video.volume(0);
48   catnews_video.hide();
49
50   video = createCapture(VIDEO);
51   video.size(750, 600);
52   video.hide();
53
54   poseNet = ml5.poseNet(video, modelLoaded);
55   poseNet.on('pose', getPose);
56   console.log(ml5);
57
58
59   catnews_video.onended(changeScene);
60
61
62
63 }
```

Code – getPose(poses) – shivering

Let pose values

Detect shivering

If hands are waving horizontally, motion = ‘frozen’

```
function getPose(poses) {
  if (poses.length > 0) {
    lhx = poses[0].pose.keypoints[9].position.x;
    rhx = poses[0].pose.keypoints[10].position.x;
    lhy = poses[0].pose.keypoints[9].position.y;
    rhy = poses[0].pose.keypoints[10].position.y;


    lear = poses[0].pose.keypoints[3].position.y;
    rear = poses[0].pose.keypoints[4].position.y;

    // shivering motion
    if (prelhx != null && prerhx != null) {
      let lhMotion = lhx - prelhx;
      let rhMotion = rhx - prerhx;

      if ((lhMotion < -5 && rhMotion > 5) || (lhMotion > 5 && rhMotion < -5)) {
        // if hands are waving horizontally
        frozen_timer++;
        if (frozen_timer > 1) {
          motion = 'frozen'; // '얼어붙은'
        }
        else motion = null;
      }
      else frozen_timer = 0;
    }
```

Code – getPose(poses) – meow

Detect meow

If hands locate upper than ears,
motion = ‘meow’

```
    // cat motion
    if (lhy <= lear && rhy <= rear) {
      // if hand locates upper than ear
      cat_timer++;
      if (cat_timer > 1) {
          motion = 'meow'; // '고양이가'
      }
      else motion = null;
    }
    else {
      cat_timer = 0;
    }
```

Code – getPose(poses) – walking

Detect walking

If hands are waving vertically,
motion = ‘walking’

```
    // walking motion
    if (prelhy != null && prerhy != null) {
      let lvMotion = lhy - prelhy;
      let rvMotion = rhy - prerhy;

      if ((lvMotion < -5 && rvMotion > 5) || (lvMotion > 5 && rvMotion < -5)) {
        // if hands are waving vertically
        walking_timer++;
        if (walking_timer > 1) {
          motion = 'walking'; // '걸어다닙니다'
        }
        else motion = null;
      }
      else walking_timer = 0;
    }
```

Code – Scene0

Title text

Start button

Game Start

If user pressed ‘Start’ button, change scene0 to scene1

```
function draw() {
  if (scene0 == true) {
    background('#D8EFD3');
    textFont(titlefont);
    textSize(80)
    noStroke();
    fill('#0BA586');
    text('CAT NEWS\nChallenge', width/2, height/2);
    textAlign(CENTER, CENTER);

    textFont(font);
    textSize(50);
    fill('#55AD9B');
    text('Start', width/2, 520);
    textAlign(CENTER, CENTER);

    if (mouseIsPressed) {
      if (mouseX > width/2 - 50 && mouseX < width/2 + 50 && mouseY > 490 && mouseY < 550) {
        // if user pressed start text
        get_started();
      }
    }
  }

```

Code – Scene1

Catnews video play

Show camera video

```
if (scene1 == true) {

  background('#D8EFD3');

  catnews_video.play();
  catnews_video.volume(1);

  image(catnews_video, -100, 0);

  videoTime = catnews_video.time().toFixed(2); // video time


  push();
  // to flip camera horizontally
  translate(width, 0);
  scale(-1, 1);
  image(video, -350, 0); // draw video of camera
  pop();
```

Text motion

If there is motion, show in the center of camera video

```
  if (motion != null) {
    text(motion, 600, 300);
    textAlign(CENTER, CENTER);
  }
```

Code – Scene1

Set value of ‘l’

Change score

If user acted right at right timing, score++

Show score

```
    if (videoTime < 3.17) l = 0;
    else if (videoTime >= 3.17 && videoTime < 6.34) l = 1;
    else if (videoTime >= 6.34 && videoTime < 9.51) l = 2;
    else l = 3;

    if (videoTime >= 0.15 + (3.17*l) && videoTime <= 0.30 + (3.17*l)) {
      if (motion == 'frozen') {
        score++;
      }
    }

    if (videoTime >= 1.23 + (3.17*l) && videoTime <= 2.10 + (3.17*l)) {
      if (motion == 'meow') {
        score += 2;
      }
    }

    if (videoTime >= 2.11 + (3.17*l) && videoTime <= 3.16 + (3.17*l)) {
      if (motion == 'walking') {
        score++;
      }
    }

    text(str(score), width/2, height/2);
  }
```

Code – Scene2

Hide videos

Show score

```
if (scene2 == true) {
  background('#D8EFD3');
  catnews_video.hide();
  video.hide();

  textFont(font);
  text('Your score is', width/2, height/2 - 100);
  textAlign(CENTER, CENTER);
  textFont(titlefont);
  text(str(score), width/2, height/2);
  textAlign(CENTER, CENTER);


}
```